

DR1/DR2 BAO Cross-check Burns the Locked Modulation and Identifies a New Natural-constant Lock That Survives Without Closing LVC v15 — Working Paper

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May 2026

Abstract

This v15 working paper reports the outcome of a series of stress tests applied to the C1 locked modulation announced as the v14 main result, and documents one new candidate lock identified during the same investigation.

The C1 lock $z_c = 1/\sqrt{3} + 1/(3\pi)$, $w = 2/(3\pi)$ is preferred over Λ CDM at $\Delta\text{BIC} = -11.38$ on the v14 baseline (Pantheon+ unbinned + Planck distance priors + SH0ES H_0 + DESI DR2 BAO; $N = 1727$). When DESI DR2 BAO is replaced by DESI DR1 BAO with the rest of the dataset unchanged, the same lock gives $\Delta\text{BIC} = +0.61$ on $N = 1725$. The result is in the BIC-tie/loss regime by the working group’s standing rule that $\Delta\text{BIC} \geq 0$ counts as loss; we therefore retire C1 as a model that does not generalise across the two BAO data releases.

A separate candidate that we provisionally name D1, locked at $z_c = \sqrt{\pi}$ with $w = 1/(\pi\sqrt{\pi})$ and reported in an internal note between v14 and the present work, is not reproducible. A direct grid scan in A at the stated lock values yields $\Delta\text{BIC} = +7.42$ over Λ CDM at the global minimum $A = +0.020$. The note’s reported value $A = -0.245$ corresponds to $\chi^2 = 1615.5$, which is 9 units worse than Λ CDM. We retire D1 as a non-finding.

In the course of the same investigation, a free six-parameter Gaussian fit (H_0 , Ω_m , $\Omega_b h^2$, A , z_c , w) was run on both DR1 and DR2 baselines. DR2 returns the C1 region. DR1 returns a different location: $z_c \approx 0.568$, $w \approx 0.127$. The DR1 best-fit width matches $2/(5\pi) = 0.12732$ to better than 0.3%. Two natural-constant locks for z_c are within 1% of the DR1 best fit and indistinguishable in χ^2 : $1/\sqrt{\pi} \approx 0.564$ and $4/7 \approx 0.571$. We call these LockB and LockC.

Both locks survive every stress test we have applied. They give $\Delta\text{BIC} \in [-6.2, -3.6]$ on the two BAO sets, leave-one-BAO out passes by BIC at 22 of 23 tested points (the only fail is removal of DR1 LRG1 at $z = 0.510$, the BAO point closest to z_c), survive a hard prior $r_d = 147.05 \pm 0.3$ Mpc with ΔBIC shifting from -6.21 to -2.46 on DR2 and from -3.60 to -1.75 on DR1, and produce a $T(z)$ profile that is internally consistent (zero at $z = 0$, zero at $z \gg 1$, peak at $z = z_c$).

However, the locks do not close. The free- w leave-one-out test, with z_c held at $4/7$, shows that DR1 BAO drives w to within 1% of $2/(5\pi)$ for 9 of 11 removed points, while DR2 BAO drives w to the upper bound of the search range ($w = 0.5$) for 8 of 12 removed points. The $w = 2/(5\pi)$ value is a feature of the DR1 BAO subset; DR2 BAO is essentially indifferent to it. The two BAO releases prefer different locks — C1 in DR2 and LockB/C in DR1 — and the data does not single out one as a generalising natural-constant ansatz.

We characterise the v15 status as: the v14 C1 result is retired as a release-specific signal; the D1 candidate is retired as a non-finding; LockB and LockC survive every direct kill test but are not closed as natural-constant ansätze; the family hypothesis from v12/v13 is not supported by the data. No working ansatz with a single natural-constant lock that survives both BAO releases has been identified by this work.

1 Setup

The model class is unchanged from v10 onward: a multiplicative modulation of the Λ CDM expansion rate,

$$H(z) = H_{\Lambda\text{CDM}}(z; H_0, \Omega_m) \cdot T(z), \quad (1)$$

with $T(z) \rightarrow 1$ at $z = 0$ and $z \gg 1$, r_d derived geometrically from θ_* as in v14, and modulation amplitude A as the only new parameter beyond Λ CDM in any single locked candidate.

For this paper, the modulation is taken to be a single Gaussian,

$$T(z) = 1 + A \exp \left[- \left(\frac{z - z_c}{w} \right)^2 \right]. \quad (2)$$

Different locked candidates fix (z_c, w) to different natural-constant values; only $(H_0, \Omega_m, \Omega_b h^2, A)$ are then free ($k = 4$).

The data sets used in this paper are exactly the v14 Task 2 dataset, with the DESI DR2 BAO block optionally replaced by the DESI DR1 BAO block:

- Pantheon+ unbinned: 1701 supernovae with full STAT+SYS covariance and 77 Cepheid-host calibrators, as in v14.
- DESI DR2 BAO: 1 isotropic D_V at $z = 0.295$ and six anisotropic D_M/r_d , D_H/r_d pairs at $z \in \{0.510, 0.706, 0.934, 1.321, 1.484, 2.330\}$ (13 measurements). Or, alternatively:
- DESI DR1 BAO: 1 isotropic D_V at $z = 0.295$ and five anisotropic pairs at $z \in \{0.510, 0.706, 0.930, 1.317, 2.330\}$ (11 measurements).
- BOSS DR12: D_M , D_H at $z = 0.38$ (2 measurements).
- eBOSS DR16: D_M , D_H at $z \in \{0.698, 1.480, 2.334\}$ and D_V at $z = 0.845$ (7 measurements).
- Planck θ_* used to derive r_d in geometric fits.
- SH0ES $H_0 = 73.04 \pm 1.04$ km/s/Mpc.
- Planck 2018 compressed distance priors (R, ℓ_A, ω_b) with the multiplicative calibration applied identically to all candidates as in v14.

The Pantheon+ likelihood, distance-prior likelihood, and BIC accounting are exactly those of v14. The DR2 baseline therefore reproduces v14 Task 2 to within the published precision; the v14 Task 2 numbers ($\chi^2_{\text{LCDM}} = 1606.94$, $\chi^2_{\text{C1}} = 1588.10$, $\Delta\text{BIC} = -11.38$) are reproduced to better than 0.01 in χ^2 in the present code.

2 The C1 lock does not generalise from DR2 to DR1

The C1 lock $z_c = 1/\sqrt{3} + 1/(3\pi)$, $w = 2/(3\pi)$ is the v14 main result. It is preferred at $\Delta\text{BIC} = -11.38$ on the v14 baseline ($N = 1727$).

We replace DESI DR2 BAO with DESI DR1 BAO — 11 measurements instead of 13 — and refit. All other data are unchanged.

A direct A -grid scan with (z_c, w) held at the C1 lock confirms that the $A = +0.035$ fit is the global minimum on DR1; no second basin is present.

	χ^2	H_0	Ω_m	A	r_d
Λ CDM (DR1, $k = 3$)	1598.28	70.00	0.2812	—	145.41
C1 (DR1, $k = 4$)	1591.44	69.50	0.2860	+0.0347	145.34
$\Delta\chi^2$	−6.84				
ΔBIC	+0.61				

Table 1: C1 on DR1 BAO with the rest of the v14 Task 2 dataset. $N = 1725$. The chi-squared favours C1 by 6.84 but the BIC penalty $\ln 1725 \approx 7.45$ for the extra parameter exceeds the chi-squared gain. By the standing rule $\Delta\text{BIC} \geq 0$ counts as loss, C1 is retired here.

The chi-squared decomposition splits the -6.84 across all four data blocks: Pantheon+ contributes -1.58 , BAO contributes -6.52 , distance priors contribute -1.78 , and the SH0ES residual contributes $+3.04$ (C1 prefers $H_0 = 69.5$, further from SH0ES than Λ CDM at 70.0). The net is dominated by BAO; the BAO advantage of C1 in DR2 was ≈ -17 , reduced to ≈ -6 in DR1. This is consistent with C1 being calibrated against the specific anisotropic BAO measurements in DR2 rather than against a coherent $H(z)$ feature visible in both releases.

We retire C1.

3 The D1 candidate is not reproducible

An internal note circulated between v14 and v15 reported a candidate $T(z) = 1 + A \exp[-((z - \sqrt{\pi})/(1/(\pi\sqrt{\pi})))^2]$, abbreviated D1, with claimed result $\chi^2 = 1572.94$, $A = -0.245$, $\Delta\text{BIC} = -26.55$ over Λ CDM on the DR2 baseline ($N = 1727$).

We attempted reproduction. With the lock $z_c = \sqrt{\pi}$, $w = 1/(\pi\sqrt{\pi})$ and $(H_0, \Omega_m, \Omega_b h^2, A)$ fit, the global minimum of the chi-squared is at $A = +0.020$ with $\chi^2 = 1606.90$. That is $\Delta\chi^2 = -0.04$ from Λ CDM and $\Delta\text{BIC} = +7.42$.

A coarse A -grid scan over $A \in [-0.40, +0.10]$ at fixed lock, profiling $(H_0, \Omega_m, \Omega_b h^2)$ at each A , returns a smooth single-minimum chi-squared curve. At the claimed best-fit $A = -0.245$, $\chi^2 = 1615.5$, which is $\Delta\chi^2 = +9.6$ over Λ CDM. The note’s reported value is not present anywhere on the grid.

A scan in w at fixed $z_c = \sqrt{\pi}$ confirms there is no value of w for which $A < 0$ is preferred. The chi-squared decreases mildly as w is broadened from 0.18 to 1.2, but A remains positive throughout ($+0.02$ to $+0.04$); at no w does the locked Gaussian admit the deep negative- A solution claimed.

We have not reconstructed how the note arrived at the reported values. We note as a forensic data point that the note’s $T(z)$ profile table at $z = 1$ and $z = 2.5$ lists values ($T = 0.964$ and $T = 0.911$) that are inconsistent with $w = 1/(\pi\sqrt{\pi}) = 0.180$. With that narrow w , the Gaussian is sub-percent at both $z = 1$ and $z = 2.5$ regardless of A ; the listed values would require $w \sim 0.6$. The lock parameters and $T(z)$ profile in the note are mutually inconsistent. We do not claim a specific bug, only that the report is not reproducible from its own stated parameters and that the reported amplitude lies in a region that direct scan rules out.

We retire D1.

4 What the data points to: free-Gaussian fits on DR1 and DR2

We open the lock entirely. The free-Gaussian model

$$T(z) = 1 + A \exp \left[- \left(\frac{z - z_c}{w} \right)^2 \right], \quad (3)$$

with all six parameters ($H_0, \Omega_m, \Omega_b h^2, A, z_c, w$) free, is fit to the full v14 Task 2 dataset, separately on DR2 BAO and on DR1 BAO. Five seeds with differential evolution followed by Nelder–Mead polish are used.

	χ^2	A	z_c	w
DR2 baseline ($N = 1727$)	1588.03	+0.0439	0.6691	0.2151
DR1 baseline ($N = 1725$)	1587.21	+0.0609	0.5681	0.1273

Table 2: Free-Gaussian best fits, six parameters. DR2 returns the C1 region (compare to C1 lock $z_c = 0.6835$, $w = 0.2122$); DR1 returns a different location, narrower and at lower z . All five DR2 seeds converge to the same minimum; four of five DR1 seeds converge to the location reported, and one outlier converges to a second basin at $z_c = 0.482$, $w = 0.048$ with $\chi^2 = 1587.80$, separated by a chi-squared barrier of ~ 1.5 from the main minimum.

The two best-fit points are not consistent with a single underlying modulation. The z_c values differ by 0.10 in absolute terms (15%); the w values differ by 0.09 (40%). The amplitudes differ by 39%. In v9 we set the criterion that a 19% shift of a position parameter under data perturbation constituted catastrophic over-fitting; here the shift is 15% in z_c but 40% in w , with w exceeding that threshold.

The v12/v13 family hypothesis,

$$z_c(n) = \frac{1}{\sqrt{3}} + \frac{n}{3\pi}, \quad w(n) = \frac{n+1}{3\pi},$$

predicts $z_c(0) \approx 0.577$, $w(0) \approx 0.106$ for DR1 ($n = 0$) and $z_c(1) \approx 0.683$, $w(1) \approx 0.212$ for DR2 ($n = 1$). The DR2 free-fit best matches the $n = 1$ rung well (z_c within 2%, w within 1%). The DR1 free-fit best does not match the $n = 0$ rung: z_c is off by 1.6% (acceptable) but w is off by 20% (0.127 versus 0.106). The family hypothesis is therefore not supported.

5 Two natural-constant lock candidates near the DR1 free fit

The DR1 free-fit best $(z_c, w) = (0.5681, 0.1273)$ admits two natural constant interpretations of z_c that lie within 1%:

$$\frac{1}{\sqrt{\pi}} = 0.5642 \quad (\text{deviation } -0.68\%), \quad \frac{4}{7} = 0.5714 \quad (\text{deviation } +0.60\%),$$

and one extremely clean interpretation of w :

$$\frac{2}{5\pi} = 0.12732 \quad (\text{deviation } -0.02\%).$$

We label the resulting locks LockB ($z_c = 1/\sqrt{\pi}$, $w = 2/(5\pi)$) and LockC ($z_c = 4/7$, $w = 2/(5\pi)$). The two locks are not distinguishable in chi-squared on DR1: their full-data fits differ by $\Delta\chi^2 = 0.002$. They differ by $\Delta\chi^2 = 0.92$ on DR2, with LockC slightly preferred.

We do not regard the natural-constant identification as a parameter saving in the BIC sense: the lock was chosen after looking at the data, which by working principle (e) of the project does not save a parameter. We continue to count $k = 4$ for both locks. The natural-constant match is reported as a structural fact about where the data points, not as a prior belief.

Direct fits

Lock	dataset	k	χ^2	H_0	Ω_m	A	ΔBIC
LockB	DR2	4	1594.19	69.60	0.2846	+0.0523	−5.29
LockB	DR1	4	1587.22	69.41	0.2867	+0.0604	−3.60
LockC	DR2	4	1593.27	69.58	0.2848	+0.0549	−6.21
LockC	DR1	4	1587.22	69.41	0.2867	+0.0613	−3.60

Table 3: LockB and LockC at $k = 4$, with r_d derived from θ_* . Both pass BIC on both BAO releases.

The two locks are weaker than C1 was on DR2 (−5.3 to −6.2 versus C1’s −11.4) but unlike C1 they survive the DR1 cross-check.

Leave-one-out by BAO

LockC is run with each of the 12 DR2 BAO measurement labels removed (BGS, LRG1–3, ELG2, QSO, Lya, BOSS, eBOSS LRG, eBOSS QSO, eBOSS Lya, eBOSS ELG), and with each of the 11 DR1 BAO measurement labels removed (the same set with one DESI DR2 anisotropic pair dropped). The amplitude A is refitted at each LOO subset.

	DR2 (12 LOO)	DR1 (11 LOO)
Wins by χ^2	12/12	11/11
Wins by BIC	12/12	10/11
A range	[+0.045, +0.067]	[+0.051, +0.065]
A spread (rel. to mean)	39.2%	24.4%
Worst-case ΔBIC	−0.75 (drop LRG2)	+1.87 (drop LRG1)

Table 4: LockC leave-one-BAO-out summary. The single BIC failure on DR1 occurs when LRG1 ($z = 0.510$, the BAO point closest to $z_c = 4/7 = 0.571$) is removed: that point alone supplies enough of the $\Delta\chi^2$ that without it, the remaining BAO support is insufficient for BIC. The amplitude is uniformly positive across all 23 LOO subsets.

LockB gives the same pattern: 12/12 on DR2, 10/11 on DR1, LRG1 the single fail, A uniformly positive, spread 44.6% on DR2 and 22.6% on DR1. The two locks are statistical reflections of the same DR1 feature.

BBN constraint test

We add a Gaussian prior $r_d = 147.05 \pm 0.3$ Mpc, intended to enforce BBN-compatible early-universe physics. This is the v13 analogue of the v13 §3.15 forced- r_d test, adapted to the unbinned-Pantheon+ likelihood.

The shift is comparable to but smaller than the v13 C1 shift of +8.08 under a hard $r_d = 147.05$ enforcement. LockC is somewhat more BBN-robust than C1 was, but the absolute advantage at the BBN constraint is weak (−2.5 and −1.8).

LockC	DR2	DR1
ΔBIC unforced	−6.21	−3.60
ΔBIC forced ($r_d = 147.05$)	−2.46	−1.75
shift	+3.75	+1.85
r_d at best fit (forced)	145.83	145.86

Table 5: LockC under BBN-compatible r_d prior. The ΛCDM baseline shifts similarly under the same prior ($r_d \rightarrow 145.93$ on DR2, $r_d \rightarrow 145.90$ on DR1; ΛCDM also fails to reach 147.05 when SH0ES is included). The BBN-forced advantage of LockC is reduced but remains negative on both releases.

The free- w test: where the lock is real and where it is not

We hold $z_c = 4/7$ and let w be free, so the model has $k = 5$ parameters. The full-data best fits are

$$\text{DR2 full data: } w = 0.500 \text{ (deviation from } 2/(5\pi) : +292.7\%),$$

$$\text{DR1 full data: } w = 0.1276 \text{ (deviation from } 2/(5\pi) : +0.25\%).$$

The DR1 free- w best is essentially exactly $2/(5\pi)$. The DR2 free- w best is at the upper bound of the search interval $[0.03, 0.50]$; that is, DR2 BAO actively pushes w away from the lock.

The leave-one-out reinforces this. Of the 12 DR2 LOO subsets, w stays at the upper bound for 8, sits in the range $[0.30, 0.50]$ for two more, and is at the lock or near-lock for two ($w = 0.195$ when LRG3 is removed, $w = 0.498$ when BOSS is removed; the LRG3-removed case is the only strictly stable one). Of the 11 DR1 LOO subsets, w is within 1% of $2/(5\pi)$ for 9, and within 21% for the remaining two (LRG1 and BOSS removed).

The honest reading is that $w = 2/(5\pi)$ is a feature of DR1 BAO, not a feature shared with DR2. LockC’s $\Delta\text{BIC} = -6.21$ on DR2 is real but is achieved by enforcing $w = 2/(5\pi)$ where the DR2 data is nearly indifferent, not by capturing a structure DR2 actively prefers.

$T(z)$ profile sanity check

For LockC at $A = +0.06$:

$$T(0) = 1, \quad T(0.4) = 1.010, \quad T(0.5) = 1.044, \quad T(z_c) = 1.060, \quad T(0.7) = 1.022, \quad T(0.8) = 1.002, \quad T(z \geq 1) = 1$$

to four decimal places. The bump is localised in $z \in [0.2, 0.95]$, $T \rightarrow 1$ at both ends, and the peak is at z_c . This is internally consistent with the lock parameters, in contrast to the D1 profile.

6 Status

The two surviving candidates are not killed by any test we have run: not by Pantheon+ unbinned, not by Planck distance priors, not by SH0ES H_0 , not by leave-one-BAO-out at 22 of 23 tested points, and not by a hard prior $r_d = 147.05 \pm 0.3$ Mpc. Their $T(z)$ profiles are internally consistent. Their natural-constant matches are tighter than those of C1: $w = 2/(5\pi)$ matches the DR1 free-fit w to 0.02% in relative deviation.

They are not closed. They differ in z_c at the percent level and the data does not single out either; $w = 2/(5\pi)$ is supported by DR1 but not by DR2; the natural-constant identification was made after the fact; the bump amplitude varies by 25–45% across LOO subsets; one DR1

Candidate	Status	Reason
v6 dual-field	retired (v7)	lost on data
v7 wavelet (z_c free)	retired (v8.5)	broke under eBOSS DR16 + PP unbinned
v9 R6 ($w = 2$)	retired (v10)	superseded
v10 R7 ($n = 0$ family)	not directly reverified here	—
v12/v13 C1 ($n = 1$ family)	retired (v15)	DR1 cross-check $\Delta\text{BIC} = +0.61$
v14-note D1 ($z_c = \sqrt{\pi}$)	retired (v15)	not reproducible from its own parameters
LockB ($z_c = 1/\sqrt{\pi}$, $w = 2/(5\pi)$)	live, not closed	survives all kill tests; w lock is DR1-only
LockC ($z_c = 4/7$, $w = 2/(5\pi)$)	live, not closed	survives all kill tests; w lock is DR1-only

Table 6: Candidate status as of v15.

LOO point (LRG1) fails BIC. The conclusion that there is a single global modulation visible to the program in the current dataset is not warranted.

What v15 establishes that was not established in v14:

1. The v14 main result C1 is release-specific: it fails on DR1.
2. The v12/v13 family hypothesis is not supported: DR1 free-fit w deviates from the predicted $n = 0$ value by 20%, while z_c is in the right neighbourhood.
3. The DR1 BAO subset, when freed, points to a clean natural-constant lock $w = 2/(5\pi)$ that is not pointed to by DR2.
4. The D1 candidate from the post-v14 internal note is not reproducible from its stated parameters.

7 Limitations

1. The DR1 BAO data files used here carry the same compilation that the previous session used; we have not re-extracted them from the DESI DR1 release with full systematics. Any error there would affect the LockB/LockC numerical values but not the qualitative finding that the two BAO releases pull in different directions.
2. No full Boltzmann Planck likelihood is used. We carry forward the compressed distance prior calibration from v14. The $\sim 0.5\%$ calibration offset on ℓ_A is independent of $T(z)$ at the sound-horizon integration scale and cancels in ΔBIC .
3. No Lagrangian formulation. The model name LVC remains aspirational.
4. Family hypothesis not supported but not affirmatively replaced. The two surviving candidates do not form a family under any rule we have checked.
5. One DR1 LOO point fails BIC for both LockB and LockC, namely removal of LRG1 at $z = 0.510$ (the BAO point closest to z_c). This is consistent with the LockB/LockC bump being supported predominantly by one data point, which is a robustness concern.
6. DR3 is the next test. If the DR3 BAO best fits remain near the DR1 location, $w = 2/(5\pi)$ becomes a serious candidate; if they migrate back to the C1 location, the program is back to a release-specific fluctuation of unknown origin.

8 Reproducibility

The numerical results in this paper are reproduced by `lvc_v15_reproduce.py`. It builds on the v14 reproduction script, with the additions:

- DESI DR1 BAO block as an alternative to DR2.
- Free-Gaussian fitter (six parameters) with multi-seed differential evolution and Nelder–Mead polish.
- Locked-Gaussian fitter for arbitrary (z_c, w) .
- DR1-aware leave-one-out for 11 BAO labels (the v14 LOO covered DR2 labels only).
- Free- w leave-one-out at $z_c = 4/7$.
- BBN $r_d = 147.05 \pm 0.3$ Mpc prior switch.

Tabulated χ^2 values reproduce within ≤ 0.05 and BIC values within ≤ 0.10 . Multi-seed fitting introduces small (< 0.05 in χ^2) variation across runs. Total runtime to reproduce all v15 tables on a single CPU core is approximately 90–120 minutes, dominated by the LOO sweeps.

Note on scope

This v15 paper is a working document. It retires two candidates (C1, D1) and keeps two open (LockB, LockC). It does not introduce a new family hypothesis, does not derive an action, and does not claim that LockB or LockC is the correct ansatz. It establishes that the v14 result was release-specific and that the program currently has no single-lock ansätze that survives both BAO releases as a closed natural-constant candidate. The next decisive input is DESI DR3.